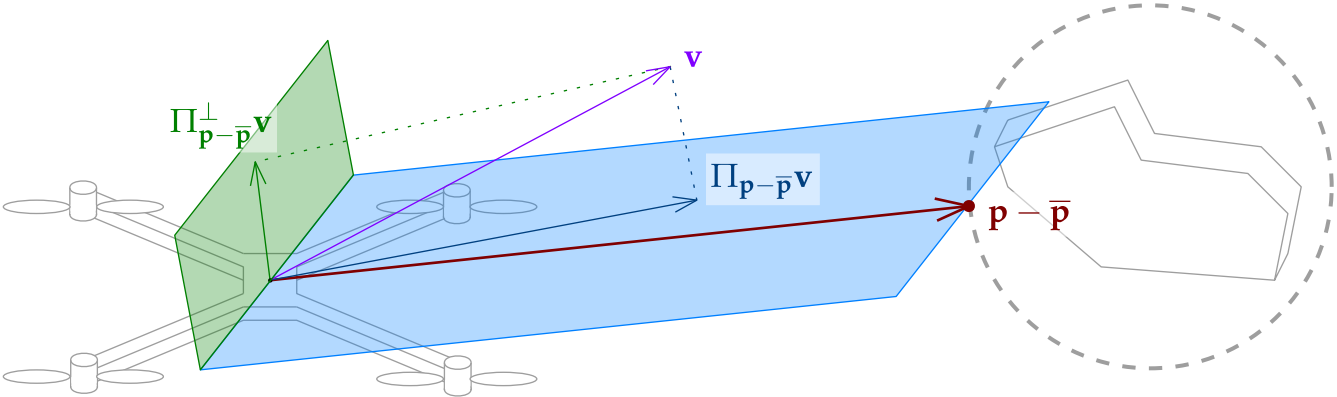
$\mathbf{v}$
$\Pi^{\perp}_{\mathbf{p}-\bar{\mathbf{p}}}\mathbf{v}$
$\Pi_{\mathbf{p}-\bar{\mathbf{p}}}\mathbf{v}$
$\mathbf{p}-\bar{\mathbf{p}}$